

Replication Code for: “A Joint Model for Multivariate Longitudinal Biomarkers and Survival Outcomes with a Fast Kalman Filter”

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Data: End-stage renal disease (dialysis) cohort, n = 5,707 subjects, 48 monthly visits

Pipeline Structure at a Glance

Script	Role	Sources	Key outputs
<code>save_theta_trans.R</code>	Step 0 — one-time utility	—	<code>theta_trans_3_0504.RData</code>
<code>albumin_fit.R</code>	Main step 1 — model fit	<code>delta_llh_OU</code> , <code>delta_negllh_OU</code> , <code>filter_estimate_OU_withP</code> , <code>llh_survival_x_OU</code>	<code>theta_est_0613.RData</code> , Figure 2
<code>bootstrap_ci.R</code>	Main step 2 — bootstrap CIs	same four (inside parallel loop)	Table 2, Table S4, Figure 4 / S1
<code>albumin_plot.R</code>	Main step 3 — figures	<code>filter_estimate_OU_withP</code>	Figure 3
<code>model_comparison_synth.R</code>	Secondary — sensitivity analysis	<code>data_preparation_functions</code> , <code>helper_functions_losses</code> , <code>llh_survival_x_OU</code> , <code>filter_estimate_OU_withP</code> , <code>llh_cs_cs</code> , <code>llh_cs_local</code> , <code>llh_joint_all_models</code>	<code>fit_ou/cc/cl_synth.RData</code> , <code>model_comparison_results.RData</code> , <code>model_comparison_plots.pdf</code>

Library files (sourced, not run directly): `delta_llh_OU.R`,
`delta_negllh_OU.R`, `filter_estimate_OU_withP.R`, `llh_survival_x_OU.R`,
`llh_cs_cs.R`, `llh_cs_local.R`, `llh_joint_all_models.R`,
`data_preparation_functions.R`, `helper_functions_losses.R`.

Overview

This repository contains all code needed to reproduce the tables, figures, and sensitivity analysis results in the paper. The analysis consists of two independent pipelines:

1. **Main analysis pipeline** (`albumin_fit.R` → `bootstrap_ci.R` → `albumin_plot.R`): fits the joint model to the real dialysis data and produces all manuscript figures and tables.
2. **Sensitivity analysis pipeline** (`model_comparison_synth.R`): fits three competing model specifications on synthetic data generated from the fitted model to justify the choice of OU process for individual trajectories.

System Requirements

R version

R 4.4.1 or later recommended.

Required R packages

```
install.packages(c(
  # Rcpp compilation
  "Rcpp", "RcppArmadillo", "inline",
  # Data handling
  "data.table", "dplyr", "reshape2",
  # Optimization and smoothing
  "GA", "assist",
  # Parallel computing (bootstrap)
  "parallel", "foreach", "doParallel",
  # Visualization
  "ggplot2", "gridExtra", "ggpubr",
  # Survival analysis
  "survival", "survminer",
  # Tables
  "xtable",
  # Model comparison
  "pROC"
))
```

Compilation note

Five files (`llh_survival_x_OU.R`, `filter_estimate_OU_withP.R`, `delta_llh_OU.R`, `delta_negllh_OU.R`, `llh_joint_all_models.R`) use Rcpp/inline to compile C++ code at runtime via `source()`. Compilation takes approximately 30–60 seconds per file on first call. Requires a working C++ compiler: - **Windows**: Rtools (<https://cran.r-project.org/bin/windows/Rtools/>) - **Mac**: Xcode Command Line Tools (`xcode-select --install`) + GFortran - **Linux**: `gcc` and `gfortran` (standard system packages)

Directory Setup

Place all R scripts and RData files in a single working directory. At the top of each script, set:

```
ROOT_DIR <- "." # or the full path to your working directory
setwd(ROOT_DIR)
```

No other path changes are needed.

File Descriptions

Core likelihood and filter functions

These are sourced automatically by the analysis scripts. They do not need to be run directly.

File	Description
<code>llh_survival_x_OU.R</code>	Joint log-likelihood for the cubic spline + OU + survival model. Returns $-2 \times$ log-likelihood to be minimized by <code>optim()</code> . See Section 3.4 of the paper.
<code>filter_estimate_OU_withP.R</code>	Kalman filter and smoother. Returns population trajectory <code>f_filter</code> , individual deviations <code>b_filter</code> , and posterior covariance matrices <code>P0</code> – <code>P3</code> used for confidence interval computation. See Section 3.3.
<code>delta_llh_OU.R</code>	Computes the Rosenberg <code>delta_est</code> (population spline coefficients) and <code>neg2llh</code> for the longitudinal-only OU model. Used during parameter initialization.
<code>delta_negllh_OU.R</code>	

File	Description
	Returns <code>neg2llh</code> only. Used with <code>optim()</code> for univariate initialization.
<code>llh_cs_cs.R</code>	Longitudinal log-likelihood for the cubic spline population + cubic spline individual model (sensitivity analysis only).
<code>llh_cs_local.R</code>	Longitudinal log-likelihood for the cubic spline population + local level individual model (sensitivity analysis only).
<code>llh_joint_all_models.R</code>	Joint log-likelihoods for the cubic-cubic and cubic-local models (sensitivity analysis only).
<code>data_preparation_functions.R</code>	Helper functions for constructing data arrays from long-format synthetic data (sensitivity analysis only).
<code>helper_functions_losses.R</code>	Loss metric functions (MSE, MAE, NLL, Brier, AUC-ROC, C-index) and risk-score computation for the three-model comparison (sensitivity analysis only). Defines <code>compute_risk_scores_from_filter</code> , called by <code>model_comparison_synth.R</code> .

Analysis scripts (run in order for main pipeline)

Step 0 (optional utility): `save_theta_trans.R`

One-time helper that regenerates the starting parameter vector. Run only if `theta_trans_3_0504.RData` is missing; otherwise skip directly to Step 1.

Inputs: - `albumin_data_0504.RData`: contains `fit_joint_albumin_1` (preliminary optim object) and `theta_trans_3`

Outputs: - `theta_trans_3_0504.RData`: starting parameter vector for `albumin_fit.R` - `theta_trans_est_0504.RData`: fitted vector from the preliminary fit (not used downstream)

Step 1: `albumin_fit.R`

Fits the joint model to the real dialysis data.

Inputs: - `albumin_data_0509.RData`: preprocessed dialysis cohort data (see Data Objects section below) - `theta_trans_3_0504.RData`: starting parameter vector for optimization (log-transformed scale) - `theta_trans_1_0404.RData`: starting values for survival subvector

Outputs: - `albumin_fit.RData` : full workspace including `fit_joint_albumin_1` (optim result object) - `theta_est_0613.RData` : fitted parameter vector on natural scale - Figure 2 in manuscript: population trends with 95% bootstrap CI

Runtime: approximately 3 hours on a standard desktop (single core).

Step 2: `bootstrap_ci.R`

Computes nonparametric bootstrap confidence intervals using 114 resamples (38 batches of 3, run in parallel).

Inputs: - Same data files as `albumin_fit.R` - `theta_est_0613.RData` : point estimates from Step 1 - `bootstrap_realdata_0613.RData` : pre-computed bootstrap results (114 fitted parameter vectors)

Outputs: - `theta_result_k.RData` (per batch): bootstrap parameter vectors - Table 5 (parameter CIs) and Table 5 Sigma (covariance CIs) printed to console - Figure 4 (correlation network): requires `bootstrap_realdata_0613.RData`

Runtime: approximately 18 hours per batch (3 resamples in parallel). Pre-computed results in `bootstrap_realdata_0613.RData` allow skipping directly to the table and figure generation block (Part 2 onward).

Step 3: `albumin_plot.R`

Generates all manuscript figures.

Inputs: - `albumin_data_0509.RData` - `theta_fit_theta_result_3.RData` : bootstrap parameter list (point estimate in `[[2]]`) - `theta_fit_fit_joint_albumin_8.RData` : fitted optim object - `restandardize_for_plot_0505.RData` : biomarker means and SDs for back-transformation to original clinical units

Outputs: - `0509_6pt_3156.png` : intermediate diagnostic plot on standardized scale (not in manuscript) - `0509_6pt_130_noCI_0904.pdf` : Figure 3 in manuscript (individual trajectories + six-month-ahead mortality probability, representative patient observed to month 35)

Runtime: approximately 10 minutes.

Sensitivity analysis: `model_comparison_synth.R`

Fits three model specifications on synthetic data to justify the OU individual process.

Inputs: - `synth_data_train_cubic_OU.RData`: synthetic training data (n = 5,707) - `synth_data_test_cubic_OU.RData`: synthetic test data (n = 5,707) - `theta_trans_3_0504.RData`, `theta_trans_3_0503.RData`, `theta_trans_3_0502.RData`, `theta_trans_1_0404.RData`: initial values - Sources: `data_preparation_functions.R`, `helper_functions_losses.R`, and all five likelihood/filter files

Outputs: - `fit_ou_synth.RData`, `fit_cc_synth.RData`, `fit_cl_synth.RData`: fitted parameters for each model - `model_comparison_results.RData`: test-set log-likelihood, MSE, AUC, C-index for all three models - `master_comparison_synth.RData`: full workspace snapshot - `model_comparison_plots.pdf`: comparison bar charts

Runtime: approximately 12 hours total (OU: 3 hrs, cubic-cubic: 6 hrs, cubic-local: 3 hrs).

Data Objects

Complete RData inventory

All `.RData` / `.Rdata` files in the bundle and which scripts use them. (Extensions are case-sensitive on Mac/Linux; filenames below match the bundle exactly.)

RData file	Used by	Contains
<code>albumin_data_0504.RData</code>	<code>save_theta_trans.R</code>	<code>fit_joint_albumin_1</code> (preliminary optim), <code>theta_trans_3</code>
<code>albumin_data_0509.RData</code>	<code>albumin_plot.R</code>	<code>data_split_in</code> , <code>z_in</code> , <code>X00_cube_in</code> (all subjects)
<code>data_split_in.Rdata</code>	<code>albumin_fit.R</code> , <code>bootstrap_ci.R</code>	$q \times n_{pt} \times m$ biomarker array
<code>z_in.Rdata</code>	<code>albumin_fit.R</code> , <code>bootstrap_ci.R</code>	$m \times n_{pt}$ survival indicator matrix
<code>X00_cube_in.Rdata</code>	<code>albumin_fit.R</code> , <code>bootstrap_ci.R</code>	$m \times p \times n_{pt}$ covariate array
<code>A_list_in.Rdata</code>	<code>albumin_fit.R</code> , <code>bootstrap_ci.R</code>	<code>list[npt]</code> of alive-at-t subject indices
<code>AB_list_in.Rdata</code>	<code>albumin_fit.R</code> , <code>bootstrap_ci.R</code>	

RData file	Used by	Contains
		list[npt] of present-at-t subject indices
theta_trans_3_0504.RData	albumin_fit.R, bootstrap_ci.R, model_comparison_synth.R	starting parameter vector (final restart)
theta_trans_3_0503.RData	albumin_fit.R, bootstrap_ci.R, model_comparison_synth.R	starting parameter vector (2nd restart, blending)
theta_trans_3_0502.RData	albumin_fit.R, bootstrap_ci.R, model_comparison_synth.R	starting parameter vector (1st restart, blending)
theta_trans_1_0404.RData	albumin_fit.R, bootstrap_ci.R, model_comparison_synth.R	initial survival subvector
theta_est_0613.RData	bootstrap_ci.R	fitted parameters, natural scale, length 42
bootstrap_realdata_0613.RData	bootstrap_ci.R	114 bootstrap parameter vectors (for Figure 4 / S1)
restandardize_for_plot_0505.RData	albumin_plot.R	biomarker means and SDs for back-transformation
theta_fit_theta_result_3.RData	albumin_plot.R	theta_result list; point estimate in [[2]]
theta_fit_fit_joint_albumin_8.RData	albumin_plot.R	fit_joint_albumin optim object; \$par = transformed estimate
synth_data_train_cubic_0U.RData	model_comparison_synth.R	synthetic training cohort
synth_data_test_cubic_0U.RData	model_comparison_synth.R	held-out synthetic test cohort
model_comparison_results.RData	(pre-computed result for reviewers)	comparison_results list (llh, MSE, AUC, C-index)

Detailed object descriptions for the primary data file follow.

albumin_data_0509.RData

Contains the preprocessed dialysis cohort (n = 5,707, 48 monthly visits, 5 biomarkers).

Object	Type	Dimensions	Description
data_split_in	array	$q \times npt \times m = 5 \times 48 \times 5,707$	Biomarker observations. data_split_in[k, t, i] = biomarker k for subject i at month t. Biomarkers: ALB (albumin, g/dL), IDWG (interdialytic weight gain, %), SBP (systolic blood pressure, mmHg), NLR (neutrophil-to-lymphocyte ratio, square root), HGB (hemoglobin, g/dL).
z_in	matrix	$m \times npt = 5,707 \times 48$	Survival indicators. z_in[i, t] = 1 if subject i died at month

Object	Type	Dimensions	Description
			t, 0 if alive, NA if censored before month t.
X00_cube_in	array	$m \times p \times npt = 5,707 \times 6 \times 48$	Covariate array. X00_cube_in[i, , t] = (age, sex, EKTv, diabetes, cardiovascular disease, vintage) for subject i at month t.

Derived index lists (reconstructed in each script from `z_in`):

Object	Type	Length	Description
A_list_in	list	npt = 48	A_list_in[[t]] = integer vector of subject indices alive (not yet dead or censored) at month t. Used in Kalman filter to compute population trajectory. Notation: set $\mathcal{A}_t$ in paper.
AB_list_in	list	npt = 48	AB_list_in[[t]] = integer vector of subject indices present at month t (alive or dying at t). Used in survival likelihood. Notation: set $\mathcal{B}_t$ in paper.

theta_trans_3_0504.RData

Starting parameter vector `theta_trans_3` for joint model optimization. All parameters on log-transformed scale (see Parameter Structure below).

restandardize_for_plot_0505.RData

Biomarker-specific means (`mean_alb`, `mean_idwg`, etc.) and standard deviations (`sd_alb`, `sd_idwg`, etc.) computed from the training data. Used to back-transform standardized model outputs to original clinical units for plotting.

Parameter Vector Structure

The full parameter vector `theta_trans` (length 42 for $q = 5$, $p = 6$) is optimized on the log-transformed scale. Back-transform the first $4q$ elements via `exp()` to recover natural-scale parameters.

Index	Symbol (paper)	Description
1-5	$\log(\zeta_1, \dots, \zeta_5)$	Population cubic spline smoothing variances, one per biomarker
6-10	$\log(v_1^2, \dots, v_5^2)$	OU individual innovation variances
11-15	$\log(\xi_1, \dots, \xi_5)$	OU mean-reversion rates
16-20	$\log(d_1, \dots, d_5)$	Diagonal of Cholesky factor of cross-biomarker covariance Σ
21-30	$l_1, \dots, l_{10}$	Lower-triangular elements of Cholesky factor of Σ (not log-transformed)
31	$\log(\kappa)$	Survival baseline hazard scale
32-36	$\gamma_{21}, \dots, \gamma_{25}$	Association parameters linking individual OU state to survival log-odds
37-42	$\gamma_0, \gamma_{11}, \dots, \gamma_{16}$	Survival covariate coefficients (intercept + 6 covariates)

General formula: total length = $0.5q^2 + 4.5q + p + 1$

The Cholesky factorization gives $\Sigma = LDL^T$ where $D = \text{diag}(d_1, \dots, d_q)$ and L is lower triangular with ones on the diagonal and l_vec in the lower triangle. Back-transform: `theta_est[1:(4*q)] <- exp(theta_trans[1:(4*q)])`.

Reproducibility Notes

- Random seed:** `set.seed(240)` is set at the top of `albumin_fit.R` and `set.seed(1)` in `bootstrap_ci.R`. Bootstrap results may differ slightly across platforms due to platform-specific random number generation.
- Runtime:** Total runtime for the full pipeline (fit + bootstrap + plots) is approximately 3 hours for fitting and 18 hours per bootstrap batch. Pre-computed results (`theta_est_0613.RData` , `bootstrap_realdata_0613.RData`) allow skipping the long-running steps.

3. **Rcpp compilation:** C++ code is compiled at runtime via `inline::cxxfunction`. Compilation output is printed to the console and can be ignored. If compilation fails, verify that Rtools (Windows) or Xcode/GFortran (Mac) is correctly installed.
4. **Data availability:** The dialysis cohort data (`albumin_data_0509.RData`) is available via the UCSB Box folder shared with reviewers. Due to patient privacy, the data cannot be deposited on a public repository. The synthetic data files (`synth_data_train/test_cubic_0U.RData`) are publicly available and sufficient to reproduce the sensitivity analysis results in Section 5.